Statistics
Parameter Estimation, Hypothesis Testing, Regression

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Lecture 3

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Esimation

- Debugging a program: A program contains an unknown number of bugs, say N.
- Deliberately introduce n (known) bugs and see how many of these are caught, say n_1. Estimate N.
- Equivalent to calculating the volume of an arbitrary object N-dimensions with a membership oracle and a randomised algorithm.
- Other examples from before: Fish population in the Lake. n fish in the lake are marked. n are caught after some time and the number of marked fish is counted, say n_1. The experiment (catching the fish) is repeated K times with sample values $n_1, \ldots, n_K$. Estimate the number of fish in the Lake.

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Estimators: Point and Interval Estimate

- Consider a population. Interest is to estimate a parameter associated with the population.
- First step is to obtain n **random** samples from the population and measure the parameter of interest.
- Let X_i be the random variable denoting the possible value from the i-th sample.
- Let θ be the parameter of interest.
- Let $x = \{x_1, x_2, \ldots, x_n\}$ be values from the sample measurements
- x_i is *realisation* of random variable X_i.
- Any function of the samples x is called a *statistic*.

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Estimators: Point and Interval Estimate

- $\hat{\theta} = g(x_1, x_2, \ldots, x_n)$ is a *point estimate* of θ.
- $\left(\mathsf{E}\left(\hat{\theta}\right) - \theta\right)$ is called the **bias** of the estimate.
- $\mathsf{E}\left(\left(\hat{\theta} - \theta\right)^2\right)$ is called the mean squared error of the estimate.
- An alterntaive to a point estimate is the *interval estimate* described using $\theta_l, \theta_u, \gamma$

$$\mathsf{Prob}(\theta_l \leq \theta \leq \theta_u) \geq \gamma$$

- γ is called the *confidence level* and (θ_l, θ_u) is the the *interval estimate.*

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Recap: Sums of Independent Random Variables

- Consider the sum of n independent and identically distributed (iid) random variables—X_i, $i = 1, \ldots n$.
- Let $\mathsf{E}(\mathsf{X}_1) = \mu$ and $\mathsf{VAR}(\mathsf{X}_1) = \sigma^2$.
- Let the sample sum be $\mathsf{S}_n := \sum_{i=1}^{n} \mathsf{X}_i$.
- We know that $\mathsf{E}(\mathsf{S}_n) = n\mu$ and $\mathsf{VAR}(\mathsf{S}_n) = n\sigma^2$.
- From the Law of Large Numbers, $\frac{\mathsf{S}_n}{n} \longrightarrow \mu$,
  - Intuitively, if we take a large number of samples, the sample mean will 'approach the true mean.

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Recap: Sums of Independent Random Variables

- From Central Limit Theorem, with Z being the unit Normal,

$$\text{Prob}\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} < y\right) \rightarrow \text{Prob}(Z < y)$$

  - Crudely, as n becomes large, the distribution of the sum approaches that of the Gaussian distribution with same mean and same variance as the sum.
  - Thus for large n, the Gaussian is a good approximation for the distribution of the sum of n iid random variables.

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The Unit Normal

- Z is the zero-mean Gaussian random variable with unit variance.
- If X is a Gaussian random variable with mean μ and standard deviation σ, then $\frac{\mathsf{X}-\mu}{\sigma}$ is unit normal.
- The above operation essentially *shifts and scales* the original Gaussian, it normalises it.
- Some 'unit normal' probabilities
  - $\mathsf{Prob}(|\mathsf{Z}| < 1) \approx 0.68$
  - $\mathsf{Prob}(|\mathsf{Z}| < 2) \approx 0.95$
  - $\mathsf{Prob}(|\mathsf{Z}| < 3) \approx 0.997$
  - $\mathsf{Prob}(|\mathsf{Z}| > 1) \approx 1 - 0.68 = 0.32$

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The Unit Normal

- Prob$(0 \leq \mathsf{Z} < 1) \approx ?$
- Prob$(\mathsf{Z} \notin (0, 1)) \approx ?$
- Prob$(\mathsf{Z} \notin (0, 2)) \approx ?$
- Prob$(\mathsf{Z} > z_\alpha) = \alpha$

Percentile	90	95	97.5	99	99.5	99.9	99.95
α	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
z_α	1.28	1.645	1.96	2.33	2.58	3.08	3.27

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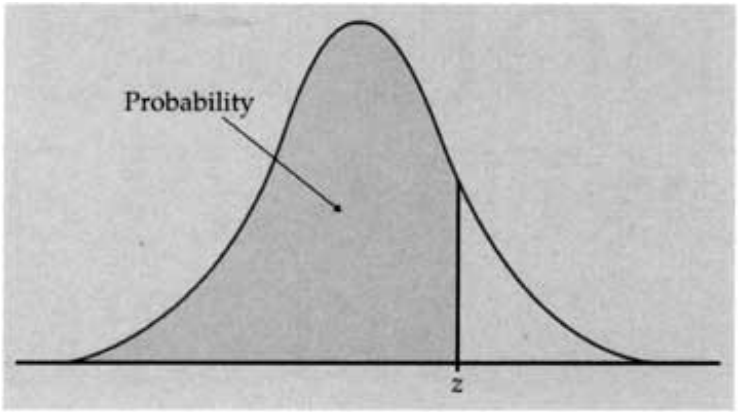


Table entry for z is the area under the standard normal curve to the left of z.

TABLE A Standard normal probabilities (*continued*)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986

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Sample Mean

- Assume a population with mean μ and variance σ^2.
  - μ is unknown and is to be estimated, σ is known.
  - Neither is known and are to be estimated.
- 'Obvious method': Obtain n independent samples whose values will be $X_1, \ldots X_n$ and calculate the sample mean as an estimator for the mean.

$$\text{Sample Mean} := \hat{\mu}_n := \frac{\sum_{i=1}^{n} X_i}{n}$$

- Clearly, $\hat{\mu}_n$ is a random variable. Characterising this random variable characterises the quality of the estimate.

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Sample Mean

- The expectation and variance of the sample mean are

$$\begin{aligned} \mathsf{E}(\hat{\mu}_n) &= \mathsf{E}\left(\frac{\sum_{i=1}^n \mathsf{X}_i}{n}\right) = \frac{\sum_{i=1}^n \mathsf{E}(\mathsf{X}_i)}{n} = \mu \\ \mathsf{VAR}(\hat{\mu}_n) &= \mathsf{VAR}\left(\frac{\sum_{i=1}^n \mathsf{X}_i}{n}\right) = \frac{\sum_{i=1}^n \mathsf{VAR}(\mathsf{X}_i)}{n} = \frac{\sigma^2}{n} \end{aligned}$$

- What is the distribution of $\hat{\mu}_n$?
- How about 'an approximate distribution' of $\hat{\mu}_n$?

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Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\begin{aligned} \mathsf{V} &:= \sum_{i=1}^{n} (\mathsf{X}_i - \mu)^2 \\ \mathsf{E}(\mathsf{V}) &= \mathsf{E}\left(\sum_{i=1}^{n} (\mathsf{X}_i - \mu)^2 \right) \\ &= \sum_{i=1}^{n} \mathsf{E}\left((\mathsf{X}_i - \mu)^2 \right) \\ &= n\sigma^2 \end{aligned}$$

- Thus if μ is known then V/n is an unbiased estimator of $\mathsf{VAR}(\mathsf{X}_1)$.

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Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$
\begin{aligned}
V &= \sum_{i=1}^{n} (X_i - \hat{\mu})^2 \\
&= \sum_{i=1}^{n} \left(X_i^2 - 2X_i\hat{\mu} + (\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} \left(X_i^2 \right) - 2\hat{\mu} \sum_{i=1}^{n} X_i + \sum_{i=1}^{n} (\hat{\mu})^2 \\
&= \sum_{i=1}^{n} \left(X_i^2 \right) - 2\hat{\mu}(n\hat{\mu}) + n\,(\hat{\mu})^2 \\
&= \sum_{i=1}^{n} \left(X_i^2 \right) - n\,(\hat{\mu})^2
\end{aligned}
$$

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Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$
\begin{aligned}
V &= \sum_{i=1}^{n} (X_i - \hat{\mu})^2 \\
\sum_{i=1}^{n} (X_i - \hat{\mu})^2 &= \sum_{i=1}^{n} X_i^2 - n\,(\hat{\mu})^2 \\
E(V) &= E\left(\sum_{i=1}^{n} X_i^2 - n\,(\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} E(X_i^2) - nE((\hat{\mu})^2) \\
&= nVAR(X_1) + n(E(X_1))^2 - nVAR(\hat{\mu}) - n(E(\hat{\mu}))^2 \\
&= n\sigma^2 + n\mu^2 - n\left(\frac{\sigma^2}{n}\right) - n\mu^2 \;=\; (n-1)\sigma^2.
\end{aligned}
$$

- Thus, when μ is not known, $\frac{V}{n-1}$ is the unbiased estimate of $VAR(X_1)$.

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Summarizing...

- Random independent samples from a population
- Each sample is an independent random variable
- Sample mean is a statistic.
- Expectation of sample mean is the same as the true mean.
- Variance of the sample mean is σ^2/n
- Since samples are independent, we can invoke LLN and CLT on the Sample Sum.

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CLT and the Sample Mean

- Let μ and σ^2 be the true mean and true variance of a population feature.
- Applying CLT to the sample sum of the n independent sample, S_n we have

$$\frac{S_n - n\mu}{\sqrt{n}\sigma} \rightarrow \mathsf{Z}$$

  where Z is the unit normal.
- Let $\mathsf{Prob}(\mathsf{Z} > z_\alpha) = \alpha$

$$\frac{S_n - n\mu}{\sqrt{n}\sigma} = n\frac{S_n/n - \mu}{\sqrt{n}\sigma} = \frac{\hat{\mu}_n - \mu}{(\sigma/\sqrt{n})}$$

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CLT and the Sample Mean

- Let us now characterize deviation of sample mean from true mean.

$$\begin{aligned}
\text{Prob}\left(-z_\alpha \leq \frac{\hat{\mu}_n - \mu}{(\sigma/\sqrt{n})} \leq z_\alpha\right) &= 1-2\alpha \\
\text{Prob}\left(-\frac{\sigma}{\sqrt{n}} z_\alpha \leq (\hat{\mu}_n - \mu) \leq z_\alpha \frac{\sigma}{\sqrt{n}}\right) &= 1-2\alpha \\
\text{Prob}\left(|\hat{\mu}_n - \mu| \leq z_\alpha \frac{\sigma}{\sqrt{n}}\right) &= 1-2\alpha \\
\text{Prob}\left(\hat{\mu}_n - \frac{\sigma}{\sqrt{n}} z_\alpha \leq \mu \leq \hat{\mu}_m + z_\alpha \frac{\sigma}{\sqrt{n}}\right) &= 1-2\alpha
\end{aligned}$$

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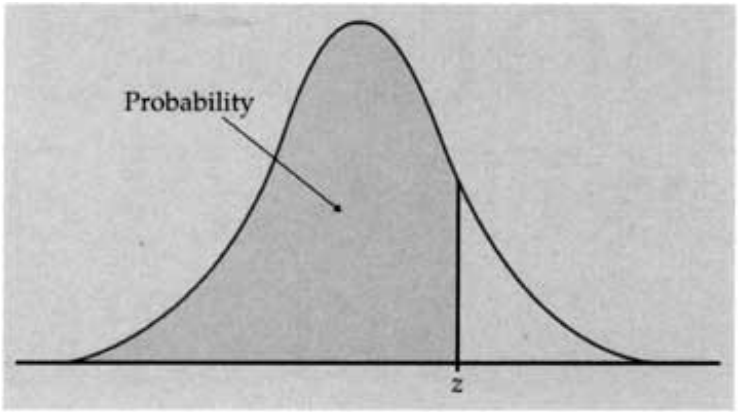


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1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
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1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
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1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
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2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986

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CLT and the Sample Mean

$$\text{Prob}\left(\hat{\mu}_n - \frac{\sigma}{\sqrt{n}}\, z_\alpha \;\leq\; \mu \;\leq\; \hat{\mu}_m + z_\alpha \frac{\sigma}{\sqrt{n}}\right) \quad = \quad 1 - 2\alpha$$

- $\left(\hat{\mu}_n - \frac{\sigma}{\sqrt{n}}\, z_\alpha, \hat{\mu}_m + z_\alpha \frac{\sigma}{\sqrt{n}}\right)$ is called the $(1-\alpha)$ confidence interval for μ.
- Usually, variance is not known! So ...
- Compute

$$\hat{V} = \frac{1}{n-1} \sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2$$

and use $\hat{V}$ as the variance, i.e., use $\sigma = \sqrt{\hat{V}}$ in the preceding.

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The $n\sigma$ rules

- 3σ and 6σ rule is something that is commonly used in quality control departments.
- Most manufacturing processes have tolerance; the allowable deviation from the nominal value.
- Production outside of this tolerance band is discarded;
- A 3σ production process means that deviation from the nominal value are such that 3σ is within the tolerance ban.
- Lower tolerance requires correspondingly lower σ and a tighter control over the process.

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Some experiments

- Write a program to do the following:
  1. Generate n IID random variables of a specified distribution F whose mean and standard deviation are known.
  2. Compute the sample mean and the limits of the 95% confidence interval.
  3. Check if the true mean is in the 95% confidence interval;
  4. In an outer loop, repeat this experiment K times.
  5. Plot the histogram of the sample mean from the K experiments.
  6. Make a scatter plot of the left and the right limits of the 95% confidence interval.

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Some experiments

- Let F correspond to Gaussian with $\mu = 0$ and $\sigma = 2$. Let $n = 20$ and $K = 1000$. Repeat for different $\sigma = 3, 5$ and $n = 50, 100, 200, 500, 1000$.
- Now do it for F uniformly distributed in $(0, 1)$, $(-1, 1)$, and $(-2, 2)$.
- And for F being a Bernouilli distribution
- Now pretend σ is not known and use $\hat{V}$ as above and see if there is a difference.

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Sampling from a Normal Population

- If X_i are i.i.d. Gaussian with mean μ and standard deviation σ then

$$\frac{\hat{\mu} - \mu}{\sigma/\sqrt{n}}$$

  is standard normal.
- Define

$$S^2 = \frac{\sum_{i=1}^{n}(X_i - \hat{\mu})^2}{n-1}$$

- Can show that $\hat{\mu}$ and S^2 are independent random variables.

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Known Variance

- Since $\hat{\mu}_n$ is Gaussian, we can say

$$\begin{aligned}
\text{Prob}\left(\mu - z_{\delta/2}\frac{\sigma}{\sqrt{n}} \leq \hat{\mu}_n \leq \mu + z_{\delta/2}\frac{\sigma}{\sqrt{n}}\right) &= 1 - \frac{\delta}{2} - \frac{\delta}{2} \\
\text{Prob}\left(\hat{\mu}_n - z_{\delta/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \hat{\mu}_n + z_{\delta/2}\frac{\sigma}{\sqrt{n}}\right) &= 1 - \delta = \gamma
\end{aligned}$$

- In this case we say that $(\hat{\mu}_n - z_{\delta/2}\frac{\sigma}{\sqrt{n}})$ is the γ confidence interval for μ.
- If the population is not from a normal distribution, invoke CLT to use the preceding as an approximation.
- Alternatively, use Chebyshev inequality to

$$\text{Prob}\left(\hat{\mu}_n - \frac{\sigma}{\sqrt{n\delta}} \leq \mu \leq \hat{\mu}_n + \frac{\sigma}{\sqrt{n\delta}}\right) \geq 1 - \delta = \gamma.$$

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Unknown Variance: Distributions of $\hat{\mu}$ and S^2

- X_i, $i = 1, \ldots, n$ are iid samples of Gaussian of mean μ and variance σ^2.
- Recall sample mean, $\hat{\mu}$, and sample variance, S^2, are

$$\begin{aligned} \hat{\mu} &= \frac{\sum_{i=1}^{n} X_i}{n} \\ S^2 &= \frac{\sum_{i=1}^{n} (X_i - \hat{\mu})^2}{n-1} \end{aligned}$$

- Sum of independent Gaussian random variables is also a Gaussian.
- Thus the sample mean $\hat{\mu}$, is Gaussian with mean μ and variance σ^2/n.
- Now consider

$$\begin{aligned} \frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2 &= \frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \hat{\mu})^2 + \frac{1}{\sigma^2} n(\hat{\mu} - \mu)^2 \\ &= \frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \hat{\mu})^2 + \left(\frac{\sqrt{n}(\hat{\mu} - \mu)}{\sigma}\right)^2 \end{aligned}$$

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Chi-Square and t Distributions

- If $Z, Z_1, \ldots Z_n$ are independent unit normal random variables then
  - $Z_1^2 + Z_2^2 + \cdots + Z_n^2$ is a χ^2 with n-degrees of freedom. It has mean 0 and variance $2n$.
  - The quantity

$$\frac{Z}{\sqrt{(Z_1^2 + Z_2^2 + \cdots + Z_n^2)/n}}$$

    has t-distribution. This has mean 0 and variance $n/(n-2)$.
- LHS is χ_n^2. Second term of RHS is χ_1^2. Hence reasonable to guess that first term of RHS is χ_{n-1}^2.

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Unknown Variance: Distributions of $\hat{\mu}$ and S^2

- Can show that $\hat{\mu}$ and S^2 are independent with $\hat{\mu}$ being normal and $(n-1)S^2/\sigma^2$ being χ^2_{n-1}.
- This means $\sqrt{n}\frac{\hat{\mu}-\mu}{S}$ is Student t distribution, t_{n-1}.
- This helps in obtaining the confidence interval when the variance is unknown.

$$\text{Prob}\left(-t_{\alpha/2,n-1} < \sqrt{n}\frac{\hat{\mu}-\mu}{S} < t_{\alpha/2,n-1}\right) = 1-\alpha$$

- To obtain confidence interval for vairance note that $(n-1)S^2/\sigma^2$ is χ^2_{n-1}. Thus

$$\text{Prob}\left(\chi^2_{\alpha/2,n-1} \leq (n-1)\frac{S^2}{\sigma^2} \leq \chi^2_{\alpha/2,n-1}\right) = 1-\alpha$$

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Parameter Estimation

- Many times, we have a model for the CDF of pdf of population feature of interest.
- The model CDF will have parameters.
- We will now discuss how to determine the **best estimate** of the paramter(s).
- As before we will obtain n independent sample realizations of the parameter of interest.

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What is being estimated?



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What is being estimated?



And what is the method?

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Likelihood Function

- $x_1, \ldots x_K$ is the set of measurements.
- Assume that the parameter set is θ.
- Question: What is the *likelihood* of observing of $x_1, \ldots x_K$ if the parameter value was θ?
  - Assume the population is Bernoulli with parameter p.
  - Assume the population is Poisson with parameter λ.
  - Assume the population is uniform in the interval $(0, a)$.
  - Assume the population is Gaussian with unit variance and mean μ.
  - Assume the population is exponential with parameter μ.

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Maximum Likelihood Estimate

- Find the θ that maximises the likelihood of the observations.

$$\hat{\theta} = \arg\max_{\theta} L(x_1, \ldots, x_K|\theta)$$

- Can also maximise the Log Likelihood function.
- Likelihood and log likelihood functions and MLEs
- Bernouilli

$$\begin{aligned} L(\underline{x}; p) &= \prod_{i=1}^{K} p^{x_i}(1-p)^{1-x_i} = p^n(1-p)^{K-n} \\ \hat{p} &= \frac{\sum_{i=1}^{K} x_i}{K} \end{aligned}$$

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